

RESEARCH ARTICLE | DECEMBER 20 2013

Continuous-wave quasi-phase-matched waveguide correlated photon pair source on a III–V chip

Peyman Sarrafi; Eric Y. Zhu; Ksenia Dolgaleva; Barry M. Holmes; David C. Hutchings; J. Stewart Aitchison; Li Qian



Appl. Phys. Lett. 103, 251115 (2013)

<https://doi.org/10.1063/1.4851095>



View
Online



Export
Citation

Articles You May Be Interested In

Parasitic nonlinearities in photon pair generation via integrated spontaneous four-wave mixing: Critical problem or distraction?

Appl. Phys. Lett. (May 2013)

Invited Article: Time-bin entangled photon pairs from Bragg-reflection waveguides

APL Photonics (August 2018)

Continuous wave second-harmonic generation using domain-disordered quasi-phase matching waveguides

Appl. Phys. Lett. (April 2009)



 Zurich
Instruments

Freedom to Innovate.

The New VHFLI 200 MHz Lock-in Amplifier.

Orchestrate pulses, triggers, and acquisition as the hub of your experiment.
Discover more – run every signal analysis tool, simultaneously.

Order now

Continuous-wave quasi-phase-matched waveguide correlated photon pair source on a III–V chip

Peyman Sarrafi,^{1,a)} Eric Y. Zhu,¹ Ksenia Dolgaleva,¹ Barry M. Holmes,² David C. Hutchings,² J. Stewart Aitchison,¹ and Li Qian¹

¹*Department of Electrical and Computer Engineering, University of Toronto, 10 King's College Road, Toronto, Ontario M5S 3G4, Canada*

²*School of Engineering, University of Glasgow, Glasgow G12 8QQ, Scotland, United Kingdom*

(Received 6 August 2013; accepted 3 December 2013; published online 20 December 2013)

We report on the demonstration of correlated photon pair generation in a quasi-phase-matched superlattice GaAs/AlGaAs waveguide using a continuous-wave pump. Our photon pair source has a low noise level and achieves a high coincidence-to-accidental ratio greater than 100, which is the highest value reported in III–V chips so far. This correlated photon pair source has the potential to be monolithically integrated with on-chip pump laser sources fabricated on the same superlattice wafer structure, enabling direct correlated/entangled photon pair production from a compact electrically powered chip. © 2013 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4851095>]

Correlated photon pair sources are of primary importance in quantum communication¹ and quantum computing.² Many previous experimental demonstrations of these sources have been reported in ferroelectric crystals and waveguides,^{3,4} fibres,^{5,6} and silicon waveguides.^{7–10} These devices, however, cannot be monolithically integrated with a pump source. The III–V semiconductor AlGaAs has been shown to be the leading material candidate for nonlinear optical interactions at telecom wavelength (1.55 μm). AlGaAs-based waveguides offer the opportunity of combining well-developed fabrication technologies to incorporate laser pump source integration with comparatively large second-order nonlinearity, low linear loss, and low two-photon absorption, making it a promising candidate for highly integrated photon pair sources.

There are two main schemes to generate correlated photon pairs in nonlinear waveguides: spontaneous four-wave-mixing (SFWM), a third-order nonlinear process relying on $\chi^{(3)}$ nonlinearity; and spontaneous parametric down conversion (SPDC), also known as parametric fluorescence, based on $\chi^{(2)}$ nonlinearity. The challenges associated with the SFWM scheme include the suppression of Raman scattering noise¹¹ and the suppression of pump light due to the pump frequency and the signal/idler frequencies located all within a limited spectral range. In contrast, SPDC-based correlated photon sources emit correlated photons with frequencies around half that of the pump frequency. Therefore, the SPDC approach allows for the possibility of the same wafer providing a pump source at the fundamental bandgap, whilst exploiting the low linear and nonlinear losses associated with half-bandgap nonlinearities.

Since the cubic symmetry of most common semiconductors, including AlGaAs, ensures the lack of a natural birefringence, an artificial means for phase-matching must be devised in order to obtain efficient SPDC in the material. This has been previously achieved in AlGaAs structures using Bragg reflection waveguides^{12,13} or vertical quasi-phase matching

(QPM) by alternating layers of different aluminum fraction.¹⁴ In this Letter, we report on a low noise, high coincidence-to-accidental ratio (CAR) (>100), continuous-wave (CW) pumped correlated photon pair source based on longitudinally quasi-phase-matched GaAs/AlGaAs superlattice waveguides. CW pumping increases the integration capability, as CW lasers are much easier to be implemented on-chip compared to pulsed lasers. This work demonstrates a correlated photon pair source that has the potential to be monolithically integrated with on-chip pump laser sources fabricated on a similar superlattice wafer structure.¹⁵

The correlated photon pair source is based on a GaAs/AlGaAs superlattice waveguide, which has a relatively large second-order susceptibility $\chi^{(2)}$. Type I QPM is achieved through quantum-well intermixing (QWI),¹⁶ arranged along the propagation direction. QWI modulates the energy band gap of the superlattice by smearing the superlattice structure and, hence, periodically suppresses the magnitude of $\chi^{(2)}$ by up to 50 pm V^{-1} .¹⁷ This method of in-plane phase-matching offers a relatively low scattering loss.¹⁸ Furthermore, both the correlated photon pairs and pump correspond to fundamental, single-lobed modes of the waveguide, allowing for efficient coupling into fibres and straightforward integration with other photonic devices. Key advantages of the QWI QPM technique is that standard wafer growth techniques are employed on a regular GaAs substrate, and that the QPM period is defined post-growth such that a range of QPM periods can be realised for a range of waveguides on the same chip. QPM by ion-implantation induced QWI has been previously utilized for second harmonic generation (SHG),^{18,19} and the birefringence of the superlattice structure has been accounted for in difference frequency generation application.²⁰

The device consists of a waveguide with a $0.6\text{-}\mu\text{m}$ -thick core layer of 14:14 monolayer GaAs/ $\text{Al}_{0.85}\text{Ga}_{0.15}\text{As}$ superlattice, with buffer layers of $300\text{ nm Al}_{0.56}\text{Ga}_{0.44}\text{As}$ on both sides, and cladding layers of $800\text{ nm Al}_{0.60}\text{Ga}_{0.40}\text{As}$. There is an additional $1\text{-}\mu\text{m}$ -thick layer of $\text{Al}_{0.85}\text{Ga}_{0.15}\text{As}$ beneath the lower cladding in order to avoid field leakage into the GaAs substrate. This heterostructure ensures a low two photon

^{a)}Electronic mail: peyman.sarrafi@mail.utoronto.ca

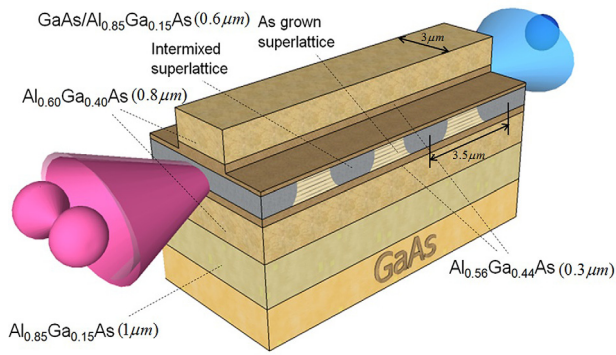


FIG. 1. A schematic of Quasi-Phase-Matched AlGaAs superlattice waveguides. Blue and pink beams are pump and generated correlated photons with a wavelength around 775 nm and 1550 nm, respectively.

absorption and good mode confinement for the 1550 nm band. A layer of PMMA was spin-coated on top of AlGaAs substrate, initially coated with a Ti-Au seed layer. Then, the gratings with different periods were defined in the PMMA mask using a Vistec VB6 electron beam lithography system. Next, a 2.3 μm -thick Au implantation mask was grown in the PMMA gaps using electroplating, followed by the removal of the PMMA mask. The ion implantation was carried out with 4 MeV As^{2+} ions at a dosage of $2 \times 10^{13} \text{ cm}^{-2}$ to create point defects in selected areas, followed by removal of the Au mask. Rapid thermal annealing (RTA) at 775 $^{\circ}\text{C}$ for 60 s provides QWI by group III atomic migration mediated by point defect diffusion. Ridge waveguides of different widths, with a height of 1 μm , were fabricated by reactive ion etching, and the resulting sample was cleaved to a length of 3.5 mm. Measurements were performed on a 3 μm -wide waveguide with a QPM periodicity of 3.5 μm . A schematic

representation of the device is depicted in Fig. 1. A scanning electron microscopy (SEM) image of the wafer structure can be found in Fig. 1 of Ref. 15.

The correlated photon pair generation experiment and coincidence count were carried out using the experimental setup depicted in Fig. 2. A TM-polarized 772.8 nm CW source was used as a pump source for producing parametrically down converted correlated photon pairs, centered around twice the pump wavelength 1545.6 nm. This light was filtered by a polarizer (passing TE polarization only), a long-pass filter, and three cascaded fiber-based pump suppression filters. The down-converted photons were then deterministically separated into two spectral bands (1562.0–1578.0 nm and 1522.0–1538.0 nm) using two fiber-pigtailed bandpass filters (BPFs) with 16.0 nm bandwidths, centered around 1570.0 and 1530.0 nm. The available BPFs are non-optimal such that they were not frequency-conjugated about the degenerate phase matching wavelength, and consequently, correlated photons can only be detected across half the bandwidth of the BPFs (1562.0–1569.2 nm and 1522.0–1529.2 nm), reducing the achievable signal to noise ratio (CAR).

Two free-running id Quantique id220 single photon detectors (SPDs), with measured quantum efficiencies of 20% at 1550.0 nm were used for coincidence measurements. The total estimated loss from the input objective to coupling-out fiber was 17 dB at 1550.0 nm. There was a 5-dB loss associated with the coupling of the pump into waveguide. The pump suppression filters had a total loss of 4.9 dB. The insertion losses of the 1570.0 and 1530.0 nm BPFs were 0.67 and 0.97 dB, respectively. Finally, an FPGA-based time interval analyzer (TIA) allowed us to measure the

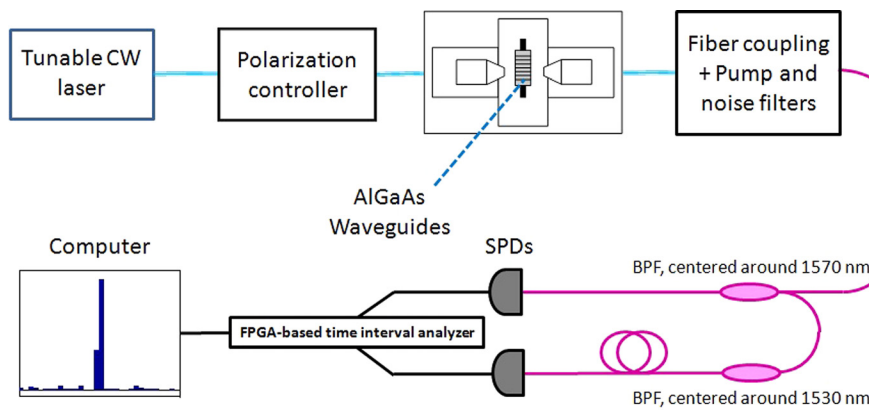


FIG. 2. Schematic of the setup for coincidence measurement in QPM AlGaAs superlattice waveguides. Blue lines are light propagating in free space, pink represents fibers.

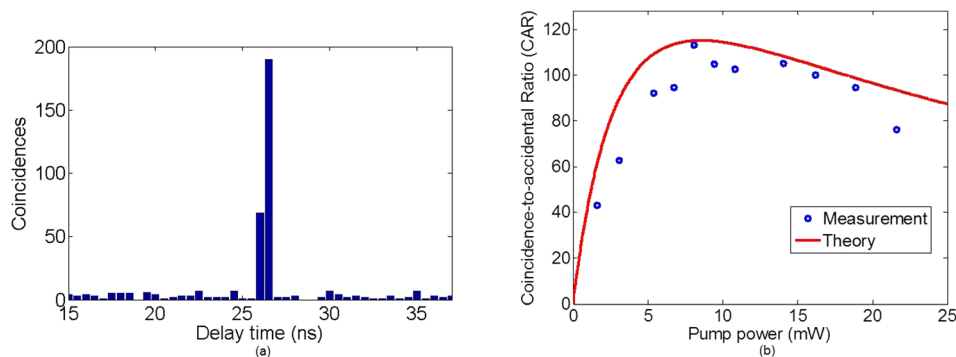


FIG. 3. (a) The typical time-bin histogram for coincidence measurements. (b) CAR as a function of coupled pump power.

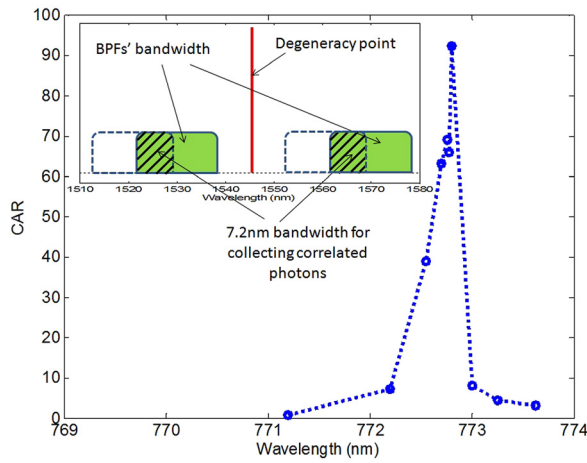


FIG. 4. CAR versus pump detuning, showing the rate of pair generation in the target spectral region versus pump wavelength. Figure inset shows the spectral coverage (in green) of the filters used for the signal and idler branches, which are not optimal because they are not symmetric with respect to the degeneracy wavelength (in red). The spectral coverage of an optimal filter (spectrally-conjugate filter) is shown as a dashed box, one for each physical filter used. Therefore, the overlap between the green and the dashed boxes (i.e., the hatched area) is the effective passband for the signal and idler photons collected after the filters.

coincidence as a function of the difference in arrival time between SPD signals.

A histogram of the raw coincidence measurement with a time-bin of 500 ps is depicted in Fig. 3(a). Here, the input CW pump power inside the sample is estimated to be ~ 8 mW, and the integration time was 200 s. Coincidence versus the difference in arrival time is plotted, and a peak at 26 ns is observed, which corresponds to the relative optical path delay of the two arms of the BPFs, demonstrating true coincidence counts due to SPDC. The finite width of the peak is due to the electronic jitter of the detection system. The total electronic jitter is estimated to be 380 ps. A detailed study of electronic jitter has shown that each detector had a 200–250 ps jitter, and there was another 200 ps jitter due to the FPGA. As shown in Fig. 3(a), true coincidence is much higher than the accidental coincidence (including the contributions from the dark counts and all other noise photons), and a high CAR (113) was measured.

In Fig. 3(b), the CAR is plotted versus pump power inside the sample (blue circles). One can observe a positive slope for the low input powers. In order to interpret this

result, we calculate the ratio of recorded true coincidences $f\alpha nl^2$ to accidental coincidences, due to single photons remaining from lost pairs and dark counts $(\alpha nl + d)^2 R$, assuming that there is no fluorescence or other type of optical noise passed by the BPFs.

$$CAR = f \frac{\alpha nl^2}{(\alpha nl + d)^2 R}, \quad (1)$$

where α is the dimensionless conversion efficiency, which is the number of generated photons pairs per second (in the 7.2 nm effective bandwidth) divided by the number of pump photons per seconds, n is the number of pump photons per second (pump power divided by the pump photon energy), l is ratio of detected photons to generated photons (10^{-3} , corresponding to the overall loss coefficient), d is the dark count of the detectors (measured to be around 2000 counts per second), and R is the resolution of the TIA (500 ps). The factor of f (for our filter pass bands and pump wavelength, 0.45) accounts for the reduction in effective bandwidth for coincidence measurements due to the non-optimal spectral coverage of the BPF pairs. The pump power at which the maximum CAR occurs is used to extract a brightness (αn) of 1.87×10^6 down converted photon pairs per second collected within the 7.2-nm effective bandwidth. We remark that the measured maximum CAR (~ 113) is very close to the theoretical maximum, which is $f^{1/(4Rd)}$ from (1), without any fitting parameter, supporting our assumption that there no fluorescence or other type of optical noise passed by the BPFs.

We experimentally confirmed that the photon pairs are generated through type-I phase matching. When the output is passed through a polarizer with its axis aligned to TM-polarization, the CAR dropped to one, indicating that no true coincidence events were observed. The phase-matching bandwidth was investigated by performing the CAR measurement as a function of the pump wavelength (at a pump power of 6 mW), with the results shown in Fig. 4. Increasing the pump wavelength beyond the degeneracy point (corresponding to SHG) results in the phase-matching condition no longer being satisfied. Decreasing the pump wavelength from the degeneracy point by only around 1 nm results in a tuning of the signal and idler wavelengths of ~ 100 nm, and therefore, they no longer coincide with the effective passband of the BPFs. Both the phase-matching bandwidth and

TABLE I. Comparison with other AlGaAs based entangled photons sources.

	This work	Ref. 12	Ref. 13	Ref. 14
Structural type	Planar GaAs/AlGaAs superlattice waveguide	Bragg-reflection waveguide	Bragg-reflection waveguide	Multilayer AlGaAs ridge waveguide
CAR @ given brightness	113 @ 1.87×10^6 pairs/s Or @ 2.6×10^5 pairs/s/nm	2 @ 6.43×10^5 pairs/s And 20 @ $\sim 6.4 \times 10^4$ (a) pairs/s	Not reported (very low by inferring from supplied data)	19.25 @ 700 pairs/s Or @ 583 pairs/s/nm
Pump	CW laser source,	Pulsed laser source,	CW laser source,	Pulsed laser source,
Measurement	Coincidence measurement	Coincidence measurement	Bell state measurement	Bell state measurement
Potential for integration	High	Moderate	Moderate	Low

^aInferred from pump power scaling.

its asymmetry are in accord with the predicted tuning curves for different frequency generation reported in Ref. 20.

There is room for improving the rate of production of the correlated photon pairs. Selecting BPFs that are frequency-conjugated at double the pump wavelength would significantly improve the measured CAR. Using a 3 dB coupler instead of 1570.0 nm/1530.0 nm, BPFs resulted in about 8 times improvement in brightness of the SPDC source at the expense of lower CAR due to an increase in noise photon detection events. The number of total true coincidences per unit time can be significantly increased by reducing the overall optical losses in the system and, in particular, improving the optical coupling between the waveguide and the collection fiber.

To show the importance of our work, we compare our results with those recently reported in Refs. 12–14 in Table I. It is shown that this work offers higher CAR and brightness compared with all other photon sources in AlGaAs platform. Also potential for integration has been compared. This work has a higher potential for integration as QPM period is defined post-growth and therefore can be tailored for different pump/signal/idler wavelength combinations. In addition, the device can be integrated with its pump on the same superlattice wafer. Integration potential is lower in Bragg-reflection waveguides because the phase matching condition cannot be tailored post-growth and also because the mode shape for the pump is non-Gaussian, making it difficult to mode-match on chip. Integration potential is even lower in Ref. 14 as this work used a non-planar structure.

In conclusion, the QWI quasi-phase-matching method in a GaAs/AlGaAs superlattice waveguide has been used to implement a source of low-noise correlated pair photons with CW pumping. Our findings demonstrate the feasibility of an electrically pumped, planar integrated, high-quality, turn-key CW photon pair source.

This work was supported by NSERC (Canada) and Engineering and Physical Sciences Research Council (UK). We acknowledge ID Quantique for providing single photon detectors utilized in this work. The authors would like to thank the Ion Beam Centre, University of Surrey, Guildford,

U.K., for carrying out the ion implantation. We also acknowledge the valuable contributions by the technical support staff of the James Watt Nanofabrication Centre at the University of Glasgow, Glasgow, U.K.

- ¹N. Gisin and R. Thew, *Nat. Photonics* **1**, 165 (2007).
- ²M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge Univ. Press, Cambridge, MA, 2000).
- ³M. Fiorentino, S. Spillane, R. Beausoleil, T. Roberts, P. Battle, and M. Munro, *Opt. Express* **15**, 7479–7488 (2007).
- ⁴S. Tanzilli, H. De Riedmatten, H. Tittel, H. Zbinden, P. Baldi, M. De Micheli, D. B. Ostrowsky, and N. Gisin, *Electron Lett.* **37**(1), 26–28 (2001).
- ⁵X. Li, P. L. Voss, J. E. Sharping, and P. Kumar, *Phys. Rev. Lett.* **94**, 053601 (2005).
- ⁶E. Y. Zhu, Z. Tang, L. Qian, L. G. Helt, M. Liscidini, J. E. Sipe, C. Corbary, A. Canagasabay, M. Ibsen, and P. G. Kazansky, *Phys. Rev. Lett.* **108**, 213902 (2012).
- ⁷N. Matsuda, H. Le Jeannic, H. Fukuda, T. Tsuchizawa, W. J. Munro, K. Shimizu, K. Yamada, Y. Tokura, and H. Takesue, *Sci. Rep.* **2**, 817 (2012).
- ⁸J. Sharping, K. Lee, M. Foster, A. Turner, B. Schmidt, M. Lipson, A. Gaeta, and P. Kumar, *Opt. Express* **14**, 12388–12393 (2006).
- ⁹H. Takesue, *IEEE J. Sel. Top. Quantum Electron.* **18**(6), 1722–1732 (2012).
- ¹⁰S. Azzini, D. Grassani, M. J. Strain, M. Sorel, L. G. Helt, J. E. Sipe, M. Liscidini, M. Galli, and D. Bajoni, *Opt. Express* **20**, 23100 (2012).
- ¹¹C. Xiong, G. D. Marshall, A. Peruzzo, M. Lobino, A. S. Clark, D.-Y. Choi, S. J. Madden, C. M. Natarajan, M. G. Tanner, R. H. Hadfield, S. N. Dorenbos, T. Zijlstra, V. Zwiller, M. G. Thompson, J. G. Rarity, M. J. Steel, B. Luther-Davies, B. J. Eggleton, and J. L. O'Brien, *Appl. Phys. Lett.* **98**, 051101 (2011).
- ¹²R. Horn, P. Abolghasem, B. J. Bijlani, D. Kang, A. S. Helmy, and G. Weihs, *Phys. Rev. Lett.* **108**, 153605 (2012).
- ¹³A. Valles, M. Hendrych, J. Svozilik, R. Machulka, P. Abolghasem, D. Kang, B. J. Bijlani, A. S. Helmy, and J. P. Torres, *Opt. Express* **21**, 10841 (2013).
- ¹⁴A. Orioux, A. Eckstein, A. Lemaître, P. Filloux, I. Favero, G. Leo, T. Coudreau, A. Keller, P. Milman, and S. Ducci, *Phys. Rev. Lett.* **110**, 160502 (2013).
- ¹⁵U. Younis, B. M. Holmes, D. C. Hutchings, and J. S. Roberts, *IEEE Photon. Technol. Lett.* **22**(18), 1358–1360 (2010).
- ¹⁶J. H. Marsh, *Semicond. Sci. Technol.* **8**, 1136 (1993).
- ¹⁷D. C. Hutchings, *IEEE J. Sel. Top. Quantum Electron.* **10**, 1124–1132 (2004).
- ¹⁸S. J. Wagner, B. M. Holmes, U. Younis, A. S. Helmy, J. S. Aitchison, and D. C. Hutchings, *Appl. Phys. Lett.* **94**(15), 151107 (2009).
- ¹⁹D. C. Hutchings, S. J. Wagner, B. M. Holmes, U. Younis, A. S. Helmy, and J. S. Aitchison, *Opt. Lett.* **35**, 1299–1301 (2010).
- ²⁰S. J. Wagner, B. M. Holmes, U. Younis, I. Sigal, A. S. Helmy, J. S. Aitchison, and D. C. Hutchings, *IEEE J. Quantum Electron.* **47**, 834–840 (2011).